

Synthetic Industrial Surface Defect Generation Under Low-Data Constraints

Evaluating Conditional GANs for Defect-Faithful Image Generation from Limited Samples

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Abstract

Industrial defect datasets are often small, highly imbalanced, expensive to collect, and difficult to share, creating a major bottleneck for computer vision systems used in automated inspection. This challenge is especially pronounced in metal-surface imagery, where rare defect cues must be learned against dominant background texture. This report investigates whether conditional generative models can synthesize useful, defect-faithful industrial surface images under such low-data constraints. The study focuses on a grayscale metal-surface dataset with three classes—normal, scratch, and spot—and uses a conditional Deep Convolutional Generative Adversarial Network (cD-CGAN) as a simple label-conditioned baseline. Rather than treating defect generation as a single monolithic problem, the work develops a staged low-data methodology that progressively simplifies, isolates, and tests defect learnability: a three-class baseline is first evaluated, then decomposed into two binary tracks (normal-vs-scratch and normal-vs-spot), followed by targeted interventions including oversampling, stratified batch sampling, defect-specific enhancement, and mild geometric augmentation. Outputs are evaluated using three qualitative criteria: visual realism, class consistency, and diversity. Results show that the model learns shared surface texture more readily than defect-specific structure; localized spot defects are easier to model than elongated scratch defects; and low-data interventions improve class consistency more clearly than diversity. The main contribution of this work is a structured experimental framework for studying synthetic industrial defect generation under scarce-data conditions and for identifying the limits of simple label-conditioned GANs, thereby motivating more controllable future approaches such as spatial guidance and image-to-image translation. **Keywords:** Generative Adversarial Networks, Conditional GANs, Synthetic Defect Generation, Industrial Computer Vision, Low-Data Learning, Surface Defect Detection, Data Scarcity.

1 Introduction

In industrial inspection settings, surface defect datasets are often small, imbalanced, expensive to annotate, and difficult to acquire under realistic operating conditions [3]. These constraints create a major bottleneck for computer vision systems used in automated inspection, particularly in manufacturing domains where rare but safety-relevant defects must be detected reliably. The challenge is especially pronounced in metal-surface imagery, where the visual signal of interest may occupy only a small portion of the image while the majority of pixels represent shared background texture.

This work addresses the data side of that inspection problem. Rather than directly training a downstream detector, the study investigates whether conditional generative models can produce synthetic industrial defect images that are realistic enough, defect-faithful enough, and varied enough to eventually support inspection-model training under low-data constraints. The objective is not simply to generate visually plausible images, but to understand what kinds of defect structures are learnable when real defect data is scarce and imbalanced, which interventions improve generation quality, and which limitations remain unresolved.

Among generative approaches, conditional Generative Adversarial Networks (cGANs) provide a natural starting point because they allow image generation to be guided by a requested class label [4]. This makes them well suited for testing whether a model can respond to a requested industrial defect type, such as a scratch or a spot, within a constrained data regime. At the same time, such models are not expected to provide full spatial control over defect placement or geometry, making them useful as a baseline for identifying both learnable patterns and architectural limitations.

The study is organized around three research questions:

1. Can the model generate the requested defect type?

2. Which defect types are easier to learn?
3. What improves output quality when real defect data is scarce and imbalanced?

To answer these questions, the work begins with a label-conditioned cDCGAN baseline trained across three classes—normal, scratch, and spot. After observing limitations in the three-class setting, the task is simplified through binary decomposition into normal-vs-scratch and normal-vs-spot tracks. Several low-data interventions are then evaluated, including oversampling, stratified batch sampling, defect-specific enhancement, and mild geometric augmentation. The resulting contribution is not a production-ready defect generator, but a structured experimental framework for studying synthetic industrial defect generation under realistic data constraints and for identifying the limits that motivate more controllable future methods.

2 Related Work

Generative models have become an important tool for synthetic data creation when labeled samples are scarce. In computer vision, Generative Adversarial Networks (GANs) introduced adversarial learning as a way to synthesize images from latent noise by training a generator and discriminator in competition [1]. Conditional GANs (cGANs) extended this framework by incorporating side information such as class labels, allowing generation to be guided toward a requested category [4]. For image synthesis tasks, convolutional GAN variants such as DCGAN further demonstrated that deep convolutional architectures can improve training stability and visual quality relative to simpler fully connected designs [5]. Within the context of this study, a label-conditioned cDCGAN provides a natural baseline for asking whether a model can generate specific industrial defect classes under low-data constraints.

Beyond noise-to-image generation, image-to-image translation methods provide an alternative paradigm for synthetic defect creation by transforming an existing source image into a target-domain image. Supervised approaches such as pix2pix learn paired mappings between source and target images [2], while CycleGAN enables unpaired translation through cycle-consistency constraints [7]. These methods are highly relevant to industrial defect synthesis because they can preserve realistic background texture while focusing on defect insertion or transformation. However, they also introduce practical trade-offs. Paired clean-to-defect image datasets are difficult to obtain in industrial settings, making supervised translation less realistic in many applications, while unpaired translation often lacks precise control over defect placement, geometry, and severity.

More recent defect-specific synthesis work suggests that stronger controllability may require transfer-based or spa-

tially guided strategies rather than global class conditioning alone. Defect Transfer GAN, for example, was proposed to synthesize diverse industrial defects for data augmentation and is directly relevant both as a defect-specific synthesis method and as the source of the SDI dataset family used in this project [6]. Such approaches motivate the future direction of the present work: moving beyond simple label-conditioned generation toward methods that better preserve surface realism while providing more explicit control over defect structure. In this sense, the current cDCGAN experiments are positioned not as a final solution, but as a structured baseline for identifying what defect cues are learnable under scarce-data conditions and where more controllable architectures become necessary.

3 Data

The experiments use a grayscale industrial metal-surface dataset derived from the Surface Defect Inspection (SDI) dataset family, specifically a Type-A metal-surface subset [6]. The study intentionally restricts the experiments to a single surface family so that the model is not simultaneously required to learn multiple material domains. This keeps the background surface distribution relatively consistent and makes defect generation the central challenge rather than cross-domain texture variation.

The full SDI dataset contains multiple surface types and a substantially larger number of images, but the present work focuses on the Type-A subset and adopts a class distribution aligned with the baseline image-synthesis setup reported in the Defect Transfer GAN work [6]. The selected counts are:

- Normal: 702 images
- Scratch: 344 images
- Spot: 181 images

This subset choice was motivated not only by consistency with prior work, but also by the desire to preserve a realistic low-data industrial setting. In many practical inspection scenarios, it is difficult to obtain large numbers of diverse defect samples, especially for rare classes. Restricting the study to a few hundred images per class therefore provides a more realistic test of whether class-conditioned generative models can learn useful defect cues under scarcity. At the same time, the resulting class imbalance creates an additional challenge: matching minority defect classes to the larger normal class through repetition-based oversampling can improve class exposure during training, but it can also introduce bias by repeatedly presenting the same rare samples.

All images are resized to 128×128 and represented in grayscale. This image size was chosen to reduce computational load while preserving the primary visible defect cues. Even under this simplification, the setting remains

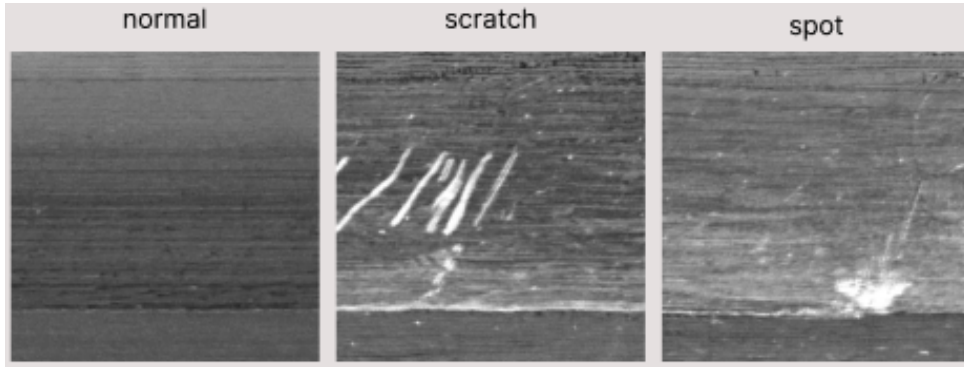


Figure 1: Representative real images from the Type-A subset of the SDI dataset used in this study. From left to right: normal, scratch, and spot. The examples illustrate both the shared metal background texture and the relatively small visual footprint of the defect signal.

difficult because only a relatively small portion of each image corresponds to the defect itself, while most pixels describe shared metal texture. In other words, the experiments operate in a high background-texture / low defect-signal regime.

The class counts further highlight the central low-data constraint of the project. The dataset is not only limited in size but also strongly imbalanced, especially for the spot class. This imbalance directly motivated the experimental design and the low-data interventions introduced later in the study, including oversampling, stratified batch sampling, and defect-specific enhancement.

4 Methods

4.1 Baseline Model Choice

The study begins with a conditional Deep Convolutional Generative Adversarial Network (cDCGAN) as a simple label-conditioned baseline. This model was selected to test whether a generator could respond to a requested defect class under scarce-data conditions. The conditioning signal is a class label $y \in \{\text{normal}, \text{scratch}, \text{spot}\}$.

Conditional GANs are a useful starting point for this problem because they allow generation to be guided by a requested category while retaining a relatively simple adversarial training framework. In the present context, this makes it possible to test whether the model can generate class-specific industrial defect imagery from limited and imbalanced data, even if full spatial control is not yet available.

4.2 Model Overview

Conceptually, the generator receives a latent noise vector z together with a class label y and produces a synthetic grayscale image. The latent input z serves as the stochastic source signal for image generation, while the class label specifies the requested defect type. The discriminator then receives either a real image with its label or a generated

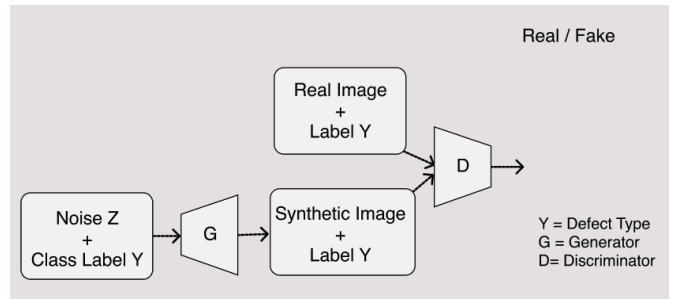


Figure 2: Simplified cDCGAN framework used in this study. The generator receives latent noise z and class label y to produce a synthetic image, while the discriminator evaluates image-label pairs as real or fake.

image with its label and predicts whether the image-label pair is real or fake. This setup makes class consistency an explicit part of the learning problem.

A simplified overview of the conditional generation process is shown in Figure 2. The figure is intended to communicate the conditioning logic of the model rather than the full layer-by-layer architecture.

4.3 Experimental Setup

The initial experiment trains a single three-class cDCGAN to generate normal, scratch, and spot images within one shared model. This serves as the baseline and reveals whether a single conditional model can learn both the shared metal texture and the class-specific defect structures.

After limitations are observed in the three-class setting, the problem is simplified through binary decomposition. Two separate models are trained:

- Normal vs Scratch
- Normal vs Spot

The purpose of this decomposition is to reduce conditional complexity and compare defect-specific learnability

more directly. It also allows defect-type-specific interventions to be tested in a more controlled way.

4.4 Low-Data Training Strategies

To improve performance under scarcity and imbalance, several targeted training strategies are evaluated:

- **Oversampling with repetition** to reduce the effective imbalance between defect classes.
- **Stratified batch sampling** to make class exposure more stable during training.
- **Randomized defect-specific enhancement** to emphasize defect cues in the scratch and spot settings.
- **Mild geometric augmentation** to introduce limited variation without destroying the primary defect structure.

These interventions are not treated as generic preprocessing steps; instead, they are part of the low-data methodology being evaluated. The goal is to determine whether targeted changes in data exposure and defect visibility improve generation quality under constrained training conditions.

4.5 Qualitative Evaluation Criteria

Generated outputs are evaluated using three qualitative criteria:

- **Visual realism:** whether the generated image plausibly matches the metal-surface domain.
- **Class consistency:** whether the generated image reflects the requested defect label.
- **Diversity:** whether outputs within the same class vary meaningfully rather than repeating similar visual structures.

These criteria are used throughout the experiments to compare the three-class baseline, the binary decomposition tracks, and the effects of the low-data interventions.

4.6 Implementation and Training Infrastructure

The training pipeline is implemented in TensorFlow/Keras and executed on the RIT Research Computing cluster. The main training environment used one NVIDIA A100 GPU, 64 GB memory, and 8 CPUs per task. Early runs encountered out-of-memory issues, which led to a refactor of the training workflow using `tf.data` and `GradientTape`. This refactor improved training stability, reduced memory pressure, and made controlled comparisons across experiments feasible.

5 Results

Generated outputs were evaluated using the qualitative criteria defined in the Methods section: visual realism, class consistency, and diversity. Since this phase of the project focused on understanding model behavior under low-data conditions rather than validating downstream task performance, evaluation was based primarily on visual inspection of representative generated outputs.

The experiments reveal four main findings.

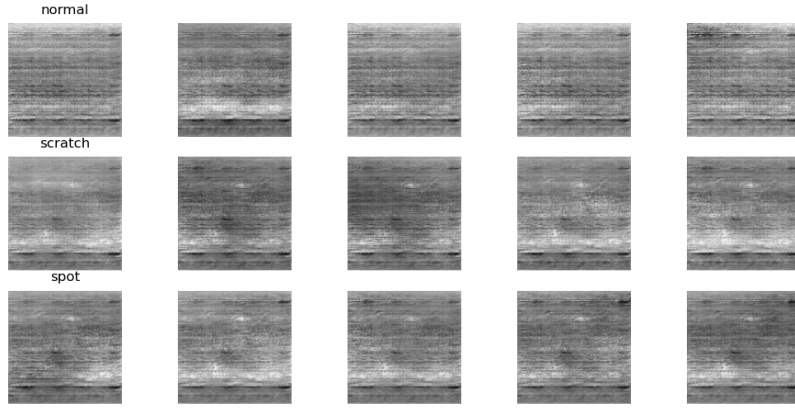
1. The three-class cDCGAN learned shared surface texture more readily than defect-specific structure. In the baseline setting, the generated outputs often resembled the overall metal-surface domain, but the scratch and spot classes were not consistently distinguishable. This suggests that the shared background texture was easier for the model to learn than the relatively small and localized class-defining defect cues.

2. Binary decomposition improved class separation. When the problem was simplified into separate normal-vs-scratch and normal-vs-spot tracks, the generated outputs showed clearer separation between normal and defect classes. Reducing the conditional complexity appeared to help the model focus more effectively on defect-specific structure.

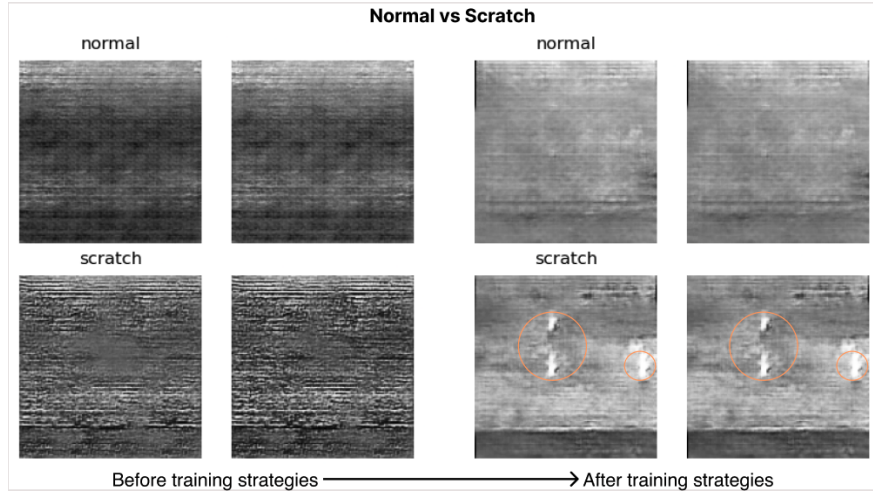
3. Defect geometry affected learnability. The spot class was easier to model than the scratch class. Localized spot-like defects were learned more reliably, whereas scratches remained more difficult because they require coherent spatial structure, orientation, and placement across the image. Even when scratch outputs became more class-consistent, they often remained structurally weak relative to real examples.

4. Targeted low-data interventions improved class consistency more than diversity. Oversampling, stratified batch sampling, defect-specific enhancement, and mild augmentation improved the visibility of class cues, especially in the binary settings. The most noticeable gains were in class consistency, particularly for the spot track. However, diversity remained limited across experiments, and outputs frequently repeated similar visual patterns.

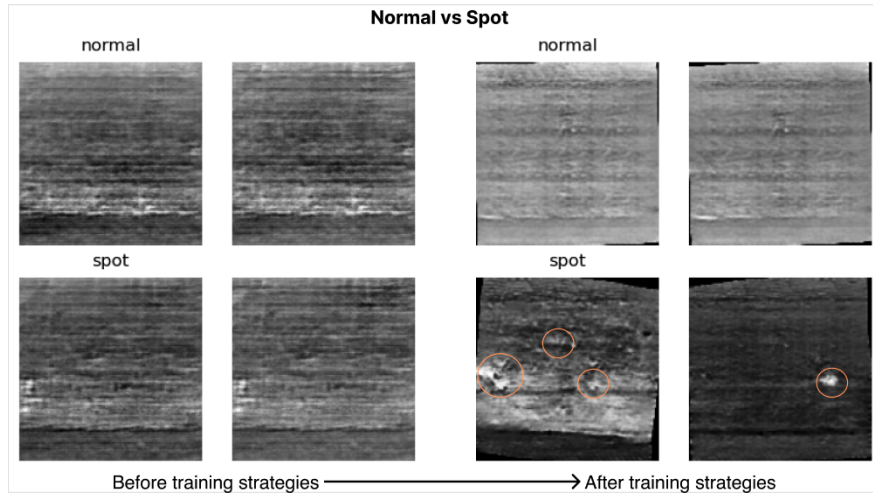
Figure 3 summarizes the visual progression across the three-class baseline and the two binary experiment tracks. Table 1 summarizes the overall qualitative outcomes.



(a) Three-class cDCGAN baseline. Five generated samples are shown for each class (normal, scratch, and spot). The outputs show that the model learned shared metal-surface texture more readily than clear class-specific defect structure, with limited within-class diversity.



(b) Normal-vs-scratch progression before and after targeted low-data interventions. Highlighted regions indicate emerging scratch-like cues. The interventions improved class separation, but scratch structure remained limited and repetitive.



(c) Normal-vs-spot progression before and after targeted low-data interventions. Highlighted regions indicate localized spot-like defects. The spot class showed the clearest gains in class consistency, although output diversity remained limited.

Figure 3: Visual progression across the main experiment tracks. Top: three-class baseline outputs. Middle: normal-vs-scratch progression before and after targeted interventions. Bottom: normal-vs-spot progression before and after targeted interventions. Across the experiments, binary decomposition and targeted low-data strategies improved class consistency more clearly than diversity.

Overall, the results indicate that the staged experimental framework was effective for diagnosing what the model could learn under low-data constraints, even though output diversity remained a persistent limitation.

6 Discussion

The main contribution of this work is methodological. The project does not claim that a simple cDCGAN solves industrial synthetic defect generation. Instead, it shows how a constrained experimental framework can be used to diagnose what the model learns easily, what remains difficult, and which low-data interventions are most useful under realistic industrial data limitations.

The results suggest that class-conditioned generation was a useful baseline because it directly tested whether a model could respond to requested defect labels. In that sense, the cDCGAN was appropriate as a starting point: it provided a controlled way to study whether defect-specific cues could be synthesized at all from limited and imbalanced data. At the same time, the experiments also showed the limits of simple global label conditioning. Although the model learned the overall metal-surface domain reasonably well, the conditioning signal alone was too coarse to reliably enforce structured defect geometry.

This limitation became especially clear in the contrast between scratches and spots. Spot defects were easier to model because they were more localized and required less coherent spatial structure across the image. Scratch defects, by contrast, depended on more extended geometry, including plausible placement, continuity, and orientation. As a result, the model often produced scratch-like contrast or texture cues without fully learning the structure of realistic scratches. This suggests that defect learnability depends not only on class identity, but also on the spatial complexity of the target pattern.

The experiments also highlight an important distinction between class consistency and diversity. Binary decomposition and targeted low-data interventions improved class separation, particularly in the normal-vs-spot setting, but these gains did not translate into strong variation within each class. Many outputs remained repetitive even when they became more visibly defect-like. This observation is important because synthetic data that is class-consistent but highly repetitive may have limited value for downstream augmentation. In other words, better conditional alignment did not automatically imply broader generative usefulness.

A further constraint was the narrowness of the dataset itself. Restricting the study to one surface family was useful for controlled experimentation because it reduced cross-domain material variation, but it also limited the range of texture and defect contexts available during training. The need to rely on repetition-based oversampling further reinforced this narrowness. This likely contributed to the limited diversity observed across outputs and suggests that

Track	Realism	Consistency	Diversity
3-class baseline	Medium	Low	Low
Normal vs Scratch	Medium	Medium	Low
Normal vs Spot	Medium	High	Low

Table 1: Qualitative evaluation summary based on visual inspection of representative generated outputs.

future experiments should compare oversampling-based balancing with alternative strategies such as normal-class downsampling or broader surface inclusion.

Taken together, these findings support the idea that simple label-conditioned GANs are valuable for diagnosing what can be learned under scarce-data conditions, but they are unlikely to provide sufficient control for structured industrial defect synthesis on their own. This directly motivates the next phase of the work: exploring more controllable approaches such as image-to-image translation [2, 7], defect-transfer methods [6], and spatially guided synthesis strategies that can preserve realistic metal texture while providing stronger control over defect structure.

7 Conclusion

This report studied synthetic industrial surface defect generation under low-data constraints using a conditional GAN baseline. The work was motivated by the need to understand whether defect-faithful synthetic images can be generated when real industrial defect data is scarce, imbalanced, and difficult to share.

Across experiments, a three-class cDCGAN baseline was compared against binary decomposition tracks and several targeted low-data interventions. The results showed that the model learned shared metal texture more readily than defect-specific structure, that localized spot defects were easier to model than elongated scratch defects, and that interventions such as oversampling, stratified batch sampling, enhancement, and mild augmentation improved class consistency more clearly than diversity.

The broader outcome of this work is a structured experimental framework for evaluating synthetic defect generation under realistic industrial data constraints. Although the present cDCGAN did not provide sufficiently diverse or fully controllable defect synthesis, it served as a useful baseline for identifying what types of cues are learnable and where stronger generative control becomes necessary. These findings provide a clear foundation for future work in image-to-image translation, segmentation-guided synthesis, and other more controllable defect-generation approaches.

8 Ethics and Limitations Statement

This project is limited by both data scope and evaluation scope. The dataset is small, imbalanced, and restricted

to a single metal-surface family, which constrains the diversity of both training inputs and generated outputs. Evaluation in the current phase is primarily qualitative, relying on visual inspection of realism, class consistency, and diversity. While this was appropriate for early-stage model understanding, stronger quantitative validation would be needed before making claims about downstream utility.

There are also broader risks in synthetic industrial data generation. Synthetic images may create a false sense of coverage if they are treated as replacements for real inspection data rather than controlled supplements. Poorly controlled generation may reinforce biases already present in the original dataset, overrepresent repetitive visual patterns, or produce unrealistic defect cues that could mislead downstream inspection models. For these reasons, future work should treat synthetic data as an augmentation tool to be validated through task-based evaluation rather than visual plausibility alone.

Future Work

The next phase of this project is motivated directly by the limitations observed in the present experiments.

- **Image-to-image translation for defect insertion:** explore whether translation-based approaches can preserve realistic metal texture while introducing more structured defect patterns than noise-to-image generation [2, 7].
- **Defect-transfer and spatial guidance strategies:** investigate whether transfer-based, mask-guided, or segmentation-guided methods can provide stronger control over defect location, geometry, and extent [6].
- **Alternative balancing strategies under low-data constraints:** compare repetition-based oversampling with alternatives such as normal-class downsampling or broader surface inclusion to better understand the trade-off between class balance and diversity.
- **Stronger evaluation methods:** extend the current qualitative framework with class-specific validation, diversity-oriented measures, and downstream inspection-model testing to assess whether synthetic images improve practical task performance [3].

Code Availability

The codebase developed for this project is being organized for public release and will be made available at: <https://github.com/SrujanVaddiparthi/DSCI601-generative-ai-for-sustainability.git>

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